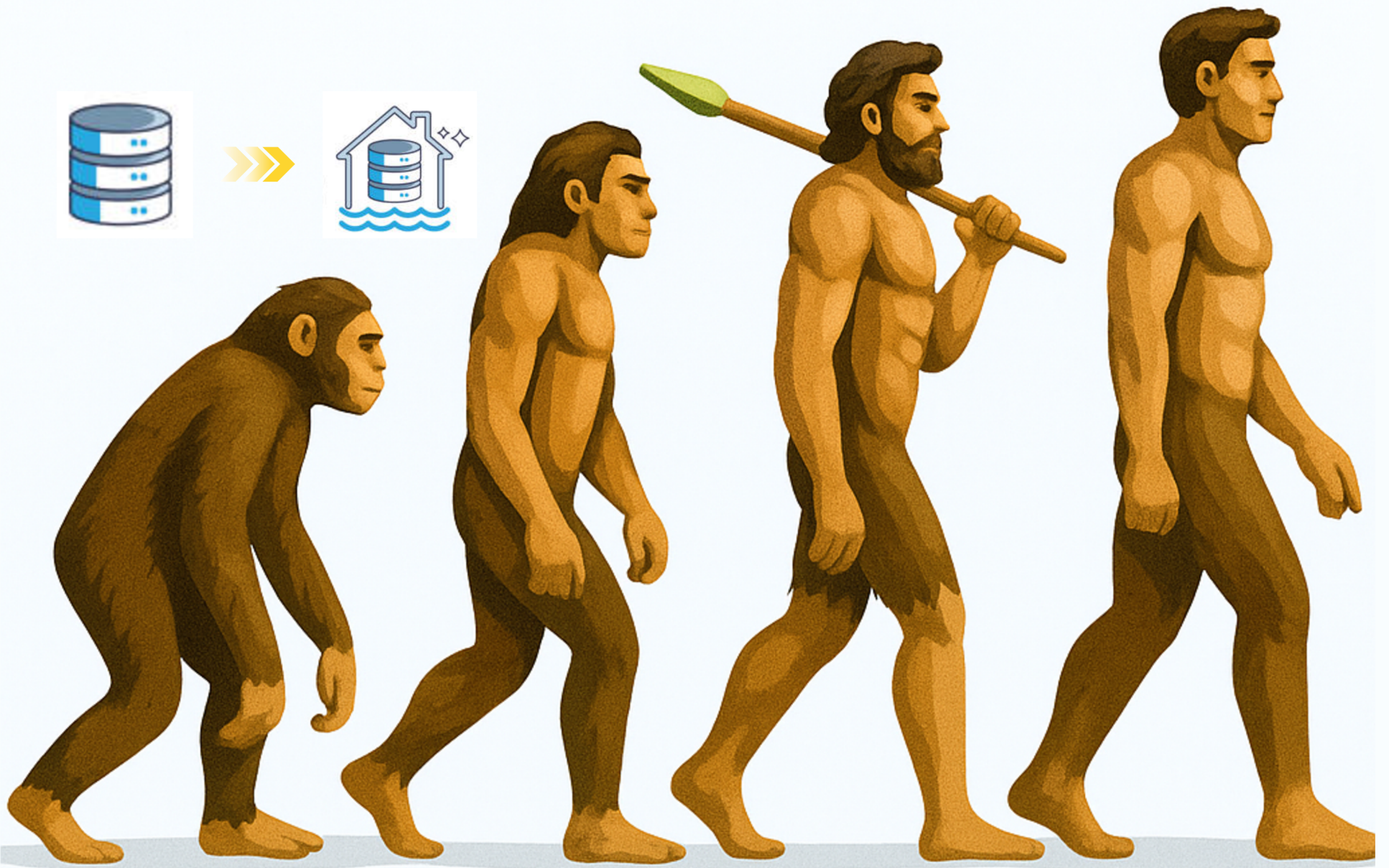


EVOLUTION OF DATA STORAGE



FROM
DATABASE TO LAKEHOUSE

swipe left





Relational Database

Store and manage transactional data.

Use Case: Rapid data entry/retrieval (e.g., banking, e-commerce).

✓ **Support:**

- ACID compliance.

✗ **No support for:**

- Unstructured data
- Analytics / BI
- Big Data & Machine Learning

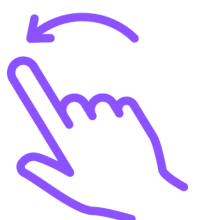


↻ Growing data size and types made RDBMS inefficient for analytics



Save it

swipe left





Data Warehouse

Store data from multiple sources for BI and reporting.

Use Case: Business intelligence, reporting, analytics (e.g., sales dashboards).

✓ Supports:

- ACID compliance.
- Complex Analytics

✗ No support for:

- Big Data
- Machine Learning



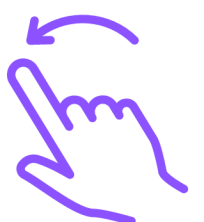
**DATA
WAREHOUSE**

🔄 Solved analytics gaps, but couldn't handle raw/unstructured data for machine learning and big data



Save it

swipe left





Data Lake

Store vast amounts of raw, unprocessed data.

Use Case: Data science, machine learning, and raw data archival (e.g., logs, videos).

✓ Supports:

- All data types to store.
- Big Data & ML



**DATA
LAKE**

✗ Lacks:

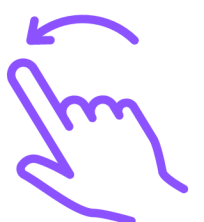
- ACID compliance.
- BI Performance

🔄 Enabled ML use cases, but not reliable for analytics



Save it

swipe left





The Split Problem

Data Warehouse

- Designed for analytics and reporting
- Supports ACID compliance & complex queries

Data Lake

- Stores all types of data (structured, semi-structured, unstructured)
- Great for big data & machine learning

The result?

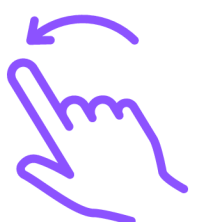
- Data is stored in separate systems
- Hard to manage,
- Expensive to maintain
- Complicated to get full insights.

 **Businesses needed ONE unified solution!**



Save it

swipe left





Data Lakehouse

Combine the capabilities of a data warehouse and the flexibility of a data lake.

Use Case: Unified analytics + data science (e.g., AI + BI in one place).

✓ Supports:

- ACID compliance.
- BI & Analytics.
- All data types
- Big Data & ML



**DATA
LAKEHOUSE**

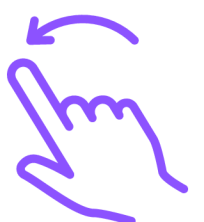
✗ Setup & tuning matters for performance.

↻ One platform to rule them all



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